

Numerical Project in Python n.2

Semidefinite embedding

Andrea Simonetto

version: January 7, 2026

We will consider the machine learning task of embedding a high-dimensional data set in the smallest dimensional space. This will be formulated as an optimization problem featuring semidefinite constraints. The goal of this project is for you to code an SDP in `cvxpy` and solve it. Incidentally, we will also determine Earth's radius. **This project counts 20% of your grading.** The project will be graded as follows: 10 points for the technical exactness; 10 points for the clarity, formatting, and readability of the report; 10 points for the graphs (font size, explanations, selections).

Some instructions.

- You can do the project in pairs (the same one of project 1), as long as everybody does a comparable amount of work.
- After the first three hours, you need to submit what you have done at that point via the Moodle page of the course. Name the file: `LastName1_LastName2_Project2_After_three_hours.py`
- By **March, Tuesday 10, at 13H00 Paris time**, you need to submit a report in a single .pdf file, of no more than 6 pages, that is readable, and it has inside all the elements to assess your work. Name the file: `LastName1_LastName2_Project2_Final.pdf`

You also need to submit a python script that we can run to assess the correctness of your plots and graphs. The script will have to generate the results that you have in the report. And it has to be readable. Name the file: `LastName1_LastName2_Project2_Final.py`

No extension allowed, No resit allowed.

- As you figured already, you will use python, and in particular you will need the following packages (please refrain from using any other packages that are not strictly needed).

```
import numpy as np
import matplotlib.pyplot as plt
import cvxpy as cp
import pickle
```

- Do as much as you can during the TP session (3 hours) at ENSTA, but it is possible that you will need to do extra work in addition to it.
- Don't wait the last day to put together the project.

Writing a readable report is very important in life. The readability of the report will be very important for the final grade, as said. Some typesetting instructions follow.

1. Save and incorporate figures and graphs in .pdf format. (No .jpeg, no .png, no print-screen). In order to help you generate readable graphs, you have access to a setup example python script. It saves a sample figure in .pdf, and so must you.
2. The labels and legends of the graph should have a fontsize similar to the one used in the main text.
3. Equations should be punctuated. They are objects living in your text, so add commas, and full-stops.
4. Don't start a sentence with a symbol.
5. Don't mix French with English.
6. Don't put code in the text, or print-screen of results. Especially with black background.

Getting started

Gauss wanted to measure the sum of the internal angles of triangles defined over the surface of Earth, to show that it was indeed greater than 180° . He wanted to achieve that by measuring pair-wise distances between known locations. We will try to do something similar with very different techniques.

You have access to `numpy` matrices collecting pair-wise distance measurements between locations in the continental USA. In particular, you can load them as,

```
with open('Distance_matrices.pkl', 'rb') as f:
    Dproto, D60, D30, D15, D = pickle.load(f)
```

You can use `Dproto` for prototyping, and when you are sure that the code is right, you can increase the dimension. `D` is the full matrix, of dimension 7800.

Each matrix D is symmetric, with zero on the diagonal. Indicate with n the dimension of D . We want now to embed the matrix into d -dimensional points, which have the lowest dimension d and represents the pair-wise measurements. To do that, we consider the following semidefinite program,

$$\min_{X \in \mathbb{S}(\mathbf{R}^n)} \quad -\text{trace}(X) \quad (1)$$

$$\text{subject to} \quad \mathbf{1}_n^\top X \mathbf{1}_n = 0 \quad (2)$$

$$X_{ii} + X_{jj} - 2X_{ij} = D_{ij} \quad \forall (i, j), i > j, \quad (3)$$

$$X \succeq 0. \quad (4)$$

The solution X^* will provide our embedding. In particular, let (λ_i, v_i) be its i -th eigenvalue-eigenvector pair.

The number of non-zero eigenvalues will be the number of minimal dimensions, that is d . The d -dimensional embedding is then,

$$x_i = \sqrt{\lambda_i} v_i \in \mathbf{R}^n, \quad i = 1, \dots, d.$$

That is there are n points in a d -dimensional space. The coordinate of the j -th point is $(x_{1,j}, \dots, x_{d,j})$.

Question 1 (4pt) Code the SDP into `cvxpy` and show that $d = 3$ up to small numerical precision. Use this result to argue that Earth cannot be flat, nor a cylinder. To answer this question, I have provided also useful plotting and postprocessing routines, see `Embedding_main.py`.

Question 2 (4pt) Try your code with different D , increasing the dimension of the SDP problem. `cvxpy` will not manage to scale to capture the full D . Start by sparsifying the constraints, meaning considering only

$$X_{ii} + X_{jj} - 2X_{ij} = D_{ij} \quad \forall (i, j) \in \mathcal{S}, i > j,$$

for a properly defined $\mathcal{S}$. In particular, you could consider to add the constraint only if

```
np.random.binomial(size=1, n=1, p= 0.2)
```

is 1. In this way, we have less constraints and `cvxpy` is likely to manage a bit better. Give the expression of the computational complexity of an interior-point method with and without the sparsification and argue if this is an effective idea or not.

Question 3 (3pt) Bonus. Code a first-order method to solve the SDP, namely a Douglas-Rachford Splitting (see Lecture notes, or Project 1). This will allow you to scale up to the full D .

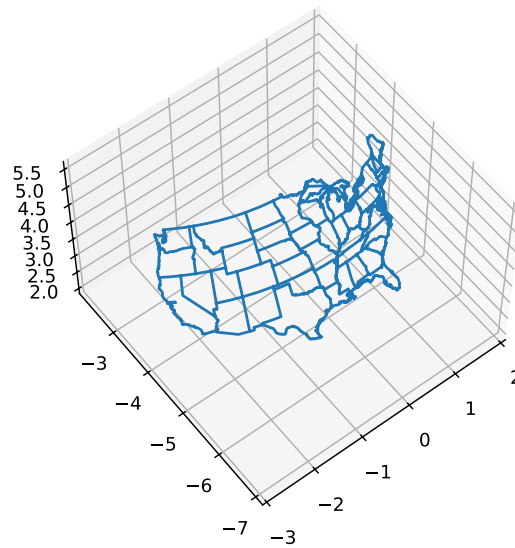


Figure 1: The data can generate the US borders onto a three-dimensional sphere. Earth is not flat and we can determine its radius by using optimization.

Earth's radius

Now that you have the three-dimensional points x , we can assume they live on the surface of a sphere. With this information, you could try to find such a sphere and in particular its radius. This will determine Earth's radius.

Question 4 (2pt) Consider the least-squares method presented in `LSFit.pdf` on Moodle. Use it to determine the radius of the fitting sphere. The unit of measurement of the result is 1000 km. Argue that this gives you the radius of Earth with very high precision.

Total: /10 Points.